

Recreational Marijuana Laws and Substantiated Child Maltreatment*

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Abstract

As more states adopt recreational marijuana laws (RMLs), concerns remain about how expanded cannabis access may affect child welfare. This study presents new causal evidence on the relationship between RMLs and substantiated cases of child neglect. Leveraging a generalized difference-in-differences framework that exploits the staggered rollout of RMLs across states, I first show that legalization accompanied by recreational dispensary openings increases the rate of adult marijuana use in treated states relative to control states. I then find that RML adoption with dispensary access leads to an increase of approximately two cases of substantiated child maltreatment per 1,000 children, primarily driven by increased neglect. Supplementary evidence from time-use data suggests that reductions in parental caregiving time may help explain this pattern. These findings suggest that increased child maltreatment among families exposed to expanded cannabis access may be an unintended consequence of marijuana legalization.

Keywords: Recreational Marijuana Laws, Child Maltreatment, Child Neglect

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1 Introduction

Marijuana policy in the United States has liberalized rapidly over the past decade, moving from limited medical programs to full recreational markets with retail dispensaries. As this shift has occurred, child welfare systems have had to update their framework for how parental marijuana use should factor into safety assessments, service plans, and decision-making norms. At the same time, maltreatment of children remains a high-cost social concern: experiences of neglect and abuse are associated with poorer health, weaker social development, and later mental health difficulties, even when children are not removed from the home (Hughes et al., 2017; Humphreys & Zeanah, 2015). Yet there is limited evidence on whether expanded access to cannabis for adults affects the rate at which states verify maltreatment.

Much of the emerging work on marijuana policy and child welfare (also referred to as child protection services or CPS) has looked at entries into foster care or referrals and has often found neutral or even beneficial effects. Gardner and Osei (2022), for example, report declines in rates of children entering foster care for parental drug or alcohol abuse, incarceration, physical abuse, and neglect after RMLs, suggesting that legalization may have reduced some of the system pressure to remove children from their homes. Other studies find no clear relationship between marijuana markets and reports of abuse or neglect at the community level (Freisthler & Kranich, 2022). A recent review concluded that the evidence base on legalization and child maltreatment is still thin and largely non-causal, and there is a critical lack of research on how legal cannabis affects other aspects of parenting and child adjustment (Wilson & Rhee, 2022). A common feature of the outcomes studied thus far is that they are highly sensitive to changing norms about marijuana, to reporting practices, and to agency-level tolerance. If legalization makes agencies less likely to remove children or shifts caseworkers toward in-home services, foster care entries could fall even if underlying conditions of maltreatment do not improve. This possibility highlights the need to study outcomes closer to children’s experiences in findings of substantiated abuse or neglect.

This paper addresses that gap using 2010-2022 administrative counts from the Children’s Bureau’s Child Maltreatment reports to build a state-year panel of substantiated cases per 1,000 children. The analysis exploits the staggered timing of (i) recreational marijuana laws, (ii) medical marijuana laws, and, crucially, (iii) the opening of recreational dispensaries across states to estimate the causal effect of expanded cannabis access on substantiated maltreatment. To address bias from staggered adoption, the paper implements the interaction-weighted event-study estimator of Sun and Abraham (2021), allowing separate identification of the effects of legalization and of actual retail availability, and permitting pre-trend tests.

Three main results emerge. First, consistent with the broader cannabis-policy literature, recreational laws that are followed by the opening of dispensaries produce clear increases in adult cannabis use, relative to never-treated states, conditional on other marijuana policies. Second, when states open recreational dispensaries, substantiated child maltreatment rises by approximately 1.2 additional cases per 1,000 children, implying an increase of about 14 percent. This pattern runs counter

to the more favorable results based on foster-care entries and suggests that part of the apparent benefit of legalization in that literature may reflect reduced removals rather than improved conditions in the home. Third, supplementary evidence on parental time allocation indicates that, following dispensary openings, adults in treated states reallocate time away from hands-on caregiving and toward leisure, providing a plausible mechanism consistent with neglect-driven substantiations.

The paper contributes to several strands of research. First, it provides causal evidence that recreational marijuana markets can increase child maltreatment, particularly neglect, as verified by child welfare agencies. Second, it helps reconcile prior findings of declining foster care entries after legalization with a less visible cost, by documenting increases in substantiated maltreatment that often occurs while children remain in the home. Third, it shows that treating recreational marijuana laws as a single policy can mask important heterogeneity: the most policy-relevant effects arise when legal status is paired with dispensary access. Fourth, by pairing policy variation with time-use evidence, it links a population-level legalization shock to day-to-day parenting behavior, clarifying a mechanism through which cannabis policy can affect child welfare.

2 Background and Related Literature

2.1 Marijuana Legalization and Child Welfare Involvement

Research on the relationship between recreational marijuana legalization and families' involvement with child welfare systems has identified multiple avenues for potential impacts. Increased marijuana use among parents may affect the likelihood of child abuse or neglect occurring if parental functioning is impaired (Wilson & Rhee, 2022). At the same time, child welfare organizations may adopt more tolerant policies toward marijuana itself once its use is legalized (Gardner & Osei, 2022). As a new default stance, child welfare professionals may therefore disregard marijuana use as a risk factor, also due in part to increased social acceptance of the drug (Price Wolf et al., 2019). Whereas the prior illegal status of marijuana allowed for standardized intolerance of its use, child welfare systems must now develop more nuanced abilities to assess how marijuana use impacts parental functioning and at what point children's safety and well-being are impacted. Further research is needed to inform this practice of risk assessment.

Some researchers have hypothesized that marijuana may be used to families' benefit. For example, researchers have begun to investigate whether legalized marijuana is substituted for other drugs with harsher legal penalties and other negative effects on the consumer and their families, however this has mixed evidence thus far (Ali, Mir M. et al., 2021; Gunn, 2122). Gardner and Osei (2022) found support for this hypothesis in their difference-in-differences analysis estimating the impact of RMLs on rates of entry into the foster care system. Using data from the federal Adoption and Foster Care Analysis and Reporting System (AFCARS), they demonstrated some significant declines in children entering foster care due to parental drug and alcohol abuse in RML states. They also found statistically significant decreases in children going to foster care subsequent to parental incarceration

in RML states. Since the reasons for incarceration are not specified in the AFCARS data, in this study it was not clear whether this effect is due to reductions in incarceration related to marijuana possession, other drug possession, or other factors. Rates of children entering care due to parental neglect showed significant decreases after marijuana legalization (Gardner & Osei, 2022). However, again it is not clear whether this reflects a reduction in children being removed from their families due to low tolerance for marijuana use while it had illegal status.

Some studies have found minimal impacts related to marijuana legalization. Freisthler and Kranich (2022) looked for any change in the rate of referrals for child abuse or neglect based on the density of local medical marijuana dispensaries, but found no effect. In a systematic review of the literature, Wilson and Rhee (2022) found insufficient evidence for effects of MMLs or RMLs on child abuse and neglect, and a paucity of studies on impacts to parental functioning and child adjustment. Those researchers called for more causally informative research to investigate potential harms of legalization on children.

This study adds to the existing research by clarifying the relationship between recreational marijuana access and rates of child maltreatment. By focusing on substantiated allegations of abuse or neglect rather than entries into foster care or referrals to child welfare for suspected maltreatment, this study is able to examine the impact of marijuana use on children through mechanisms related to daily care. This level of maltreatment may be less severe than foster care placement, but childhood experiences of abuse are known to have negative health impacts (Hughes et al., 2017), and there is growing evidence that chronic neglect in childhood has long-term damaging effects on mental health (Humphreys & Zeanah, 2015; Lippard & Nemeroff, 2020).

2.2 Parental Use of Recreational Marijuana

Rates of marijuana use have risen with RMLs (Anderson & Rees, 2023); evidence indicates that this includes use by parents with children under age 18. Goodwin et al. (2021) determined that RMLs lead to increased cannabis use among adults in homes with children across all sociodemographic groups, using a difference-in-difference approach. In a study on Ohio child welfare cases involving substance use before, during, and after the COVID-19 pandemic shutdowns (2019-2022), the proportion of cases involving marijuana use among parents was found to have significantly increased in 2020, despite a large decline in the overall numbers of substantiated cases compared to the pre-pandemic period (Ye et al., 2024).

This study operationalizes the availability of recreational marijuana through the opening of dispensaries, which allows for more precise analysis than law passage without increased availability of the drug. Once parents have increased access to recreational marijuana, it is feasible that some may reduce their time spent engaged in caregiving tasks. Particularly when marijuana use reaches problematic levels, this may lead to neglectful conditions for children whose needs chronically go unmet due to parental distraction or disorganization (Kepple, 2018), or abusive situations where parents have additional risk factors that interact with marijuana use or withdrawal, leading to

increased aggression (Smith et al., 2013).

2.3 Potential Mechanisms Leading to Maltreatment

Child welfare systems define and monitor multiple forms of maltreatment by caregivers, including several types of neglect and abuse. Allegations of neglect are substantiated when children’s needs go unmet to a degree that has an observed negative impact on their safety, health, or well-being (Child Welfare Information Gateway, 2025). There are several mechanisms through which marijuana use may contribute to child neglect, as shown in studies on the behavior patterns through which substance use impairs parenting capacities.

Negative impacts on parenting behaviors may be associated with two main dimensions: (1) frequency and amounts of substance use (including marijuana), and (2) severity of functional problems resulting from acute intoxication, withdrawal or recovery-related effects, and longer-term effects from chronic use (Kepple, 2018). There is evidence that parents whose level of use meets criteria for Substance Use Disorder have increased rates of neglect, due to impairments in attention, interpretation of their child’s emotional cues, cognitive functioning including problem solving and decision making, and problems with emotion regulation (Kepple, 2018). Therefore, parents with Cannabis Use Disorder may be more likely to struggle in keeping up with caregiving tasks such as keeping consistent routines for hygiene, meals, getting children to bed and to school on time; organizing medical appointments and administering treatments as prescribed; budgeting or accessing necessary resources for the household; or helping children with complex tasks such as providing support for specific mental health or developmental needs. When such parenting deficits lead to chronic neglect of children’s needs, child welfare agencies may become involved. Further research is needed to clarify the risk and extent of functional deficits resulting from excessive marijuana use.

Medical research on the health-related impacts of marijuana use supports concerns for consequent functional impairments. Regular marijuana use increases the risk of developing Cannabis Use Disorder (Hoch et al., 2025), which is characterized by cravings, tolerance, and/or withdrawal symptoms, as well as persistent use despite resulting physical, psychological, and/or social harms (Patel & Marwaha, 2024). Heavy marijuana users have increased risks for cognitive impairments, including in learning, memory, and attention (Kroon et al., 2021), as well as some serious physical health problems including stroke and cardiovascular illness (Testai et al., 2022). There is also mounting evidence that marijuana use exacerbates some mental health problems, such as anxiety, mania, psychosis, and potentially depression (Hines et al., 2020; Kroon et al., 2021; Memedovich et al., 2018; National Academies of Sciences, Engineering, and Medicine et al., 2017; Petrilli et al., 2022). All of these health consequences may contribute to challenges in daily functioning and parenting.

Physical abuse is defined as the non-accidental infliction of a physical injury by a caregiver (Child Welfare Information Gateway, 2025). Physical abuse of a child may also be influenced by marijuana use. In a meta-analysis of 30 studies, Dellazizzo et al. (2020) found evidence of a positive

relationship between violent behavior and cannabis use among youth and young adults. The authors noted that the literature suggests this may be a bidirectional causal relationship, particularly among individuals with other risk factors for violent behavior. There is evidence that marijuana withdrawal symptoms are associated with increased relationship aggression in people who have a history of aggressive behavior (Smith et al., 2013).

In a longitudinal study of individuals with diagnosed severe mental illness, the frequency and severity of marijuana use (more than once a day) were found to be significant predictors of violent behavior toward others (Beaudoin et al., 2024). In a recent systematic review of the research on the risks and benefits of regular cannabis use Hoch et al. (2025) show that there has been consistent evidence of cognitive impairments with mild to moderate effect sizes, several of which may contribute to increased rates of physical abuse among individuals who are predisposed to violent behavior. Specifically, regular cannabis users have increased risks of adverse effects in global cognition, cognitive impulsivity, decision making, and executive functioning, and additionally, mood dysregulation (Hoch et al., 2025). The reviewers further determined that there is increased risk of violent behavior among habitual cannabis users, and particularly those diagnosed with mental illness (Hoch et al., 2025). Similarly, among parents who use marijuana to a sufficient level, such effects and interactions with other risk factors may contribute to child maltreatment.

3 Data

3.1 National survey of drug use and health (NSDUH)

First, to examine changes in adult marijuana use, I use data from the 2002–2019 National Survey on Drug Use and Health (NSDUH). Although individual-level data are restricted, state-by-year averages are publicly available for this time period. The survey collects data on marijuana use in each wave, with results reported over rolling two-year intervals.

Because the NSDUH reports outcomes over overlapping two-year periods, aligning state-level policy indicators with precise policy implementation dates is not straightforward. Following the approach of Choi et al. (2019, p. 312), I define policy exposure as the share of the two-year period during which the policy was active. These exposure measures range from 0 to 1. For example, a value of 0.33 reflects implementation during the last third of the interval, or the final eight months of the two year period.

Marijuana prevalence rates are derived from self-reported data of how many days in the prior month each respondent consumed marijuana or hashish. Marijuana users are identified as those respondents reporting at least one day of marijuana use in the prior month. On average, 7.5% of adults reported prior-month use over 2002–2019. The prevalence was highest among individuals aged 18–25 at approximately 19%, compared to 5.5% among those 26 and older. Rates of use increased steadily over time, rising from 6% in 2002 to 11% by 2019.

3.2 Children’s Bureau Child Maltreatment Reports

To analyze changes in child welfare outcomes, I use data from the federal Children’s Bureau annual Child Maltreatment Reports, compiled from the administrative records submitted by state child welfare agencies to the National Child Abuse and Neglect Data System (NCANDS). These data are publicly available for all 50 states and the District of Columbia. I compiled the available datasets on Child Maltreatment from 2010 through 2022. From 2010 onward, changes in reporting requirements precipitated significant increases in rates of child maltreatment reporting from mandated professionals such as medical providers (Edwards et al., 2023). The reauthorization of the federal Child Abuse Prevention and Treatment Act (CAPTA) in 2010 and subsequent state and federal policies contributed to these increases impacting the overall observations in child welfare data (Edwards et al., 2023).

Although the Children’s Bureau does not include microdata in the Child Maltreatment Reports, its standardized structure across states enables consistent panel construction, particularly on the primary outcome used in this study, the annual rate of substantiated child maltreatment per 1,000 children in the population. Substantiations reflect cases where an investigation concluded that maltreatment occurred according to state policy definitions. The Child Maltreatment Reports also provide data on the numbers of children in each state with substantiated findings of maltreatment categorized by type, including physical abuse, neglect, sexual abuse, medical neglect, and psychological abuse. Additionally, the annual reports account the numbers of victimized children by age in each state. Because the data are annual and reflect the calendar year, policy timing is coded proportionally based on the share of the year during which a law was in effect. For instance, if a recreational marijuana law took effect on July 1 of a given year, the exposure measure for that year is coded as 0.5.

3.3 American Time Use Dataset (ATUS)

To explore mechanisms, I collect all available data from the American Time Use Dataset for years 2010-2022. The rich time dimension provides an interview date for every observation. This feature of the data permits coding the policy variables down to the day level, lending credence to their validity. The outcome variable, $Childcare_{ist}$, is derived from survey questions that ask respondents how many minutes they spend on core parenting activities.

Like the CPS, the ATUS is a nationally representative dataset. However, its statistical power is relatively limited due to the dataset’s lower size. In practice, this only hinders stratifying the data to explore heterogeneity by certain characteristics. When restricting the analysis to include only parents of children aged 0 to 17, sufficient data remain such that the law of large number is satisfied for every state in every year.

Table 1: Descriptive Statistics for Outcome Variables

	All States		Treated States		Control States	
	Mean	SD	Mean	SD	Mean	SD
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: NSDUH (2010-2019)						
<i>Used marijuana in the last 30 days</i>						
Rate, Ages 26+	6.8%	(3.1%)	10.5%	(3.4%)	6.1%	(2.5%)
State-year Observations	510		80		430	
Number of States	51		8		43	
Panel B: NCANDS (2010-2022)						
<i>Cases of Substantiated Child Neglect</i>						
Incidence (per 1000 children)	9.64	(5.36)	11.37	(5.51)	8.91	(5.13)
State-year Observations	663		260		403	
Number of States	51		20		31	
Panel C: ATUS (2010-2022)						
<i>Time Spent Caring for Household Children</i>						
Minutes per day	85.0	(26.0)	87.2	(28.0)	83.5	(24.5)
State-year Observations	663		260		403	
Number of States	51		20		31	

Notes: This table presents the average characteristics across all states (columns 1–2), treated states (columns 3–4), and control states (columns 5–6) utilized in the baseline analysis. It reports both mean values and standard deviations, with observations weighted according to relevant state population using SEER data. The initial two columns encompass all states, while columns 3–4 focus on the treated states that enacted a Recreational Marijuana Law (RML) and opened recreational dispensaries. The final two columns pertain to control states that did not open recreational dispensaries prior to the conclusion of the sample period.

3.4 Recreational Marijuana Law Dates

I use the effective dates catalogued by Anderson and Rees (2023), updated with my own research into legislative changes. The relevant dates are listed in appendix table A1.

4 Methods

I exploit variation across state and time in the enactment of recreational marijuana laws to identify their effects on substantiated child maltreatment and parental time use. Following recent developments in econometrics literature which suggest that TWFE estimates within a staggered adoption framework may be biased Goodman-Bacon, 2021, I eschew the traditional two-way fixed effect method entirely and use the heterogeneous estimator proposed by Sun and Abraham (2021) for all estimation.

Because the formal Sun and Abraham (2021) Stata package does not produce summary estimation, I follow their paper in constructing summary estimates by linearly combining all post-treatment CATT estimates. Each period is weighted by the number of states contributing to that estimate. This yields an ATT that is consistent with TWFE conceptually, but free from timing bias. Appendix table A2 summarizes the differences between this procedure and TWFE summary estimation.

4.1 Empirical model

Rather than relying on the canonical two-way fixed effects model, I estimate all effects using Abraham & Sun’s interaction-weighted estimator to account for the considerable timing heterogeneity in policy adoption throughout the sample period (Sun & Abraham, 2021). The primary model takes the following form:

$$Y_{st} = \delta_0 + \sum_{e \notin C} \sum_{\substack{l=-4 \\ l \neq -1}}^4 \delta_{e,l} (\mathbb{1}\{E_s = e\} \times RMD_{st}^l) + \delta_2 MML_{st} + \delta_3 MJDecrim_{st} + \dots$$

$$\dots + X'_{st} \delta_4 + \alpha_s + \gamma_t + \epsilon_{st} \quad (1)$$

where Y_{st} represents one of several outcomes in state s during year t . These include the prevalence of prior-month marijuana use, the rate of substantiated cases of child maltreatment per 1,000 state population, time spent by parents caring for children, and time spent by parents on leisure activities.

The regressor of interest, $RMD_{s,t}^l$, measures the share of states s which have at least one operational recreational marijuana dispensary during year t . e indexes each treated cohort by year of treatment, C is the set of all never-treated observations, and l indexes relative years to treatment for treated observations. In this case, l represents the distance in years between the current period and

the date that recreational marijuana dispensaries opened. $\mathbb{1}\{E_s = e\}$ is an indicator function where e represents each treated cohort by year of treatment in the support set of all treated cohorts E_s .

$MML_{s,t}$ and $MJDecrim_{s,t}$ measure the share of states which have enacted medical marijuana laws and marijuana decriminalization laws by time t . The recovered coefficients, the set of estimators $d_{e,l}$, therefore capture the marginal conditional average treatment effects of recreational marijuana dispensaries opening over and above the effect induced by medical law passage and marijuana decriminalization laws.

There are two additional steps to the estimation procedure. Step 2 balances the regression through estimating weights by sample share for each treated cohort. Weights are generated using the following formula:

$$Pr\{E_i = e \mid E_i \in [-l, T - l]\} \quad (1.2)$$

In step 3, the interaction weighted estimates for each period are constructed by weighting the $\delta_{e,l}$ coefficients derived from step 1 using the weights from step 2. The formula is as follows:

$$\hat{v}_g = \frac{1}{|g|} \sum_{l \in g} \sum_e \hat{\delta}_{e,l} \times \hat{Pr}\{E_i = e \mid E_i \in [-l, T - l]\} \quad (1.3)$$

where $\hat{\delta}_{e,l}$ are the unbalanced treatment effects estimated in step one. The final interaction-weighted estimates $\hat{v}_g$ for each relative time period are the reported estimates.

The vector X_{st} contains a set of time-varying, state-level controls derived from the ACS, FRED, and the CDC. They are intended to model the evolution of states' population characteristics over time, and include age, gender, race, ethnicity, education level, the real state GDP per capita, and the real minimum wage. α_s is a vector of time-invariant state effects and γ_t is a vector of state-invariant time effects.

5 Results

5.1 RMLs and marijuana use

Figure 1 and Table 2 report the first-stage relationship between recreational marijuana dispensaries opening and adult marijuana use. The event-study coefficients in the pre-treatment years show small insignificant deviations from zero, but a joint test of the leads does not reject equality to zero ($p = 0.25$). Overall, the pre-trends are consistent with the parallel-trends assumption for the NSDUH sample. After dispensaries open, prevalence of prior-month marijuana use among adults aged 26 and over increases steadily over the next two years and remains elevated for at least three more. The pattern and magnitudes are consistent with the broader legalization literature, which find large increases in self-reported marijuana use after law passage and dispensaries open (Dave et al., 2022; Sabia et al., 2024).

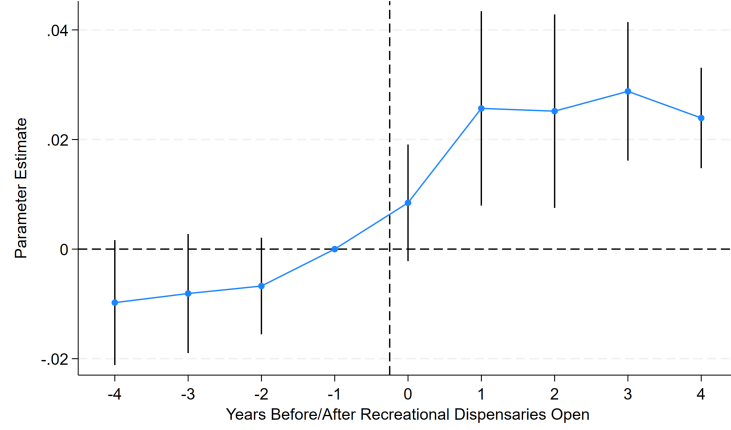
To further explore uncertainty in the pre-period, I also estimate the event study with four additional pre-treatment years. The extended-window results, shown in Appendix Figure B1, display the same qualitative pattern as Figure 1. All pre-treatment coefficients remain small and insignificant, and are jointly indistinguishable from zero.

5.2 RMLs and substantiated child maltreatment

Figure 2 plots the dynamic effects of RMLs with operating recreational dispensaries on substantiated child maltreatment per 1,000 children over 2010–2022. The pre-policy coefficients are tightly centered on zero with relatively small standard errors, and a joint test of the leads fails to reject equality to zero. There is no significant evidence of differential pre-trends between adopting and never-treated states. Starting in the year dispensaries open, the estimates turn positive and stay positive for several years, with the largest increases two to four years after market entry. The magnitudes, about 1–2 additional substantiated cases per 1,000 children, imply an overall increase of about 14 percent depending on specification.

Summary estimates are presented in Table 3. Overall I find an effect of 1.28 additional cases of substantiated child maltreatment per 1,000 children aged 0–17. This suggests that, after RMLs with operating dispensaries, child protective services more often determine that cases meet statutory abuse or neglect criteria. The result complements prior work finding flat or declining foster-care entries after legalization, by showing that verified maltreatment can rise even when removals do not.

Figure 1: Sun & Abraham Event Study, Effect of RML with recreational dispensaries on prevalence of prior-month marijuana use (NSDUH, 2010-2019)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. The regression controls for all covariates, state and time fixed effects. Standard errors are clustered at the state level. Data are from the 2010-2019 NSDUH and include all observations aged 26+.

Table 2: Summary estimates for effect of RML with dispensaries on prior-month adult marijuana use (NSDUH, 2010-2019)

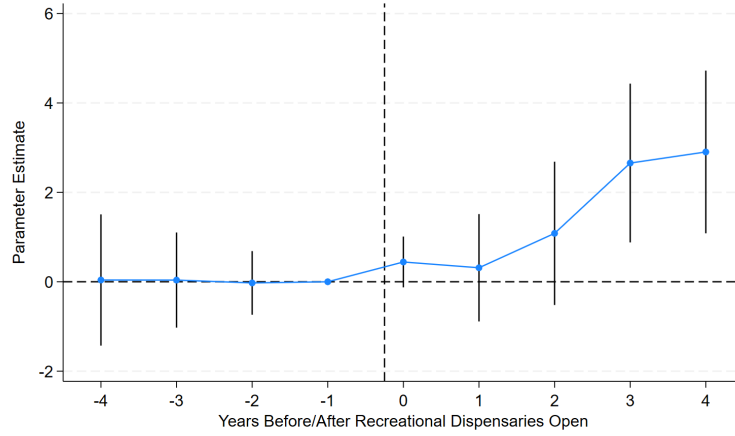
(1)	
Any Prior-month Marijuana Use	
<i>RML with dispensaries</i>	0.023*** (0.004)
F-test Leads (p-value)	0.25
Control Mean	0.061
Percent Effect	36.1%
State-year Observations	487
Model	Sun & Abraham ¹

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represents the change in the prevalence of prior-month adult marijuana use. Data are from the 2010-2019 NSDUH and include all observations aged 26+. All regressions include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

¹ Sun & Abraham point estimates are an average treatment effect derived from a weighted linear combination of estimates from all post-treatment periods. Never-treated units are used for the control group.

Figure 2: Sun & Abraham Event Study, Effect of RML with recreational dispensaries on incidence of substantiated child maltreatment (Children’s Bureau, 2010-2022)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represents the change in substantiated cases of child maltreatment per 1,000 state population of children aged 0-17. The regression controls for all covariates, state and time fixed effects. Standard errors are clustered at the state level. Data are from the 2010-2022 NCANDS and include every case of reported child maltreatment.

Table 3: Summary estimates for effect of RML with dispensaries on substantiated cases of child maltreatment per 1,000 children (Children’s Bureau, 2010-2022)

(1)	
Substantiated Cases (per 1,000)	
<i>RML with dispensaries</i>	1.28*** (0.39)
F-test Leads (p-value)	0.99
Control Mean	8.54
Percent Effect	14.2%
State-year Observations	534
Model	Sun & Abraham ¹

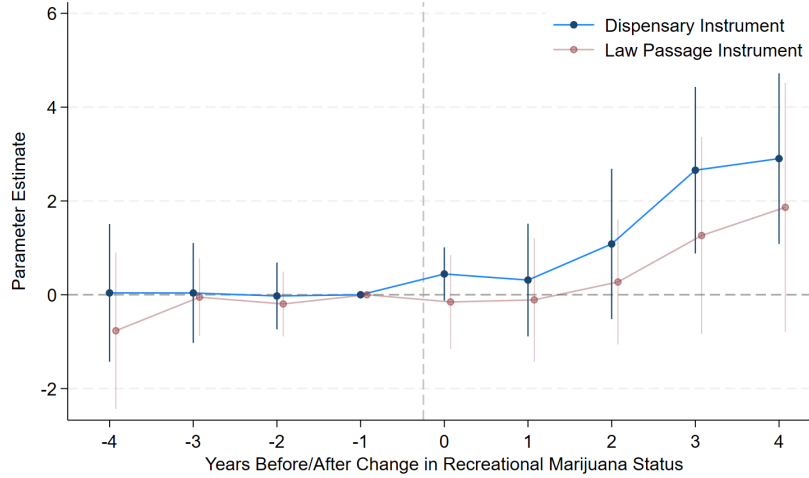
* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represents the change in substantiated cases of child maltreatment per 1,000 state population. Data from the 2010-2022 NCANDS and include all reported cases of child maltreatment. All regressions include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

¹ Sun & Abraham point estimates are an average treatment effect derived from a weighted linear combination of estimates from all post-treatment periods. Never-treated units are used for the control group.

5.3 Sensitivity to Alternative Policy Construction

Figure 3: Sun & Abraham Event Study, Effect of RML with recreational dispensaries on incidence of substantiated child maltreatment (Children’s Bureau, 2010-2022)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represents the change in substantiated cases of child maltreatment per 1,000 state population of children aged 0-17. The regression controls for all covariates, state and time fixed effects. Standard errors are clustered at the state level. Data are from the 2010-2022 NCANDS and include every case of reported child maltreatment.

In Figure 3, I evaluate whether the main result depends on how marijuana policy is coded. In the baseline model, treatment is defined as the point when a recreational law is in place *and* recreational dispensaries begin operating. In contrast, I re-estimate the same event-study specification but recode treatment to the year of law passage as summarized in Table A1. I plot both sets of coefficients on the same figure. The law-passage instrument (in red) shows no statistically significant effect on substantiated maltreatment in any period. In aggregate, the post-period coefficients imply a null effect.

This suggests that the increase in substantiated maltreatment appears only when treatment is defined to coincide with the start of retail dispensary operations. This pattern is consistent with legalization affecting child maltreatment primarily through market access rather than statutory change alone, similar to findings in prior research on the effects of medical marijuana dispensaries (Andreyeva & Ukert, 2019).

5.4 Heterogeneity by Maltreatment Type and Child Age

Table 4 decomposes the main effect by investigation disposition on maltreatment allegations. The increase in substantiated cases is concentrated in neglect (including inadequate supervision and failure to provide necessary care), with smaller and statistically insignificant changes in physical and psychological abuse. There is an apparent null effect for sexual abuse. This pattern is what would be expected if the relevant mechanisms for substantiated neglect passes through day-to-day caregiving lapses triggered by increased parental cannabis use.

Table 5 then examines effects by child age. The coefficients are significant for all age groups, with the largest proportional increase for older adolescents. This is somewhat surprising because younger children require more intensive, constant supervision and basic care; so chronic reductions in parental attentiveness or availability may be more likely to garner consequences in CPS investigations. A possible explanation is that although adult marijuana use is legal, parents are deemed neglectful by CPS when they give cannabis products to underage youths (Child Welfare Information Gateway, 2024). An increase in parents sharing marijuana with older teens may account for the observed effect.

Table 4: Estimates for effects of RML with recreational dispensaries on substantiated child maltreatment by type (Children’s Bureau, 2010-2022)

	(1)	(2)	(3)	(4)	(5)
	Neglect	Physical Abuse	Sexual Abuse	Psychological Abuse	Medical Neglect
Cases (per 1,000)					
<i>RML with dispensaries</i>	1.13*** (0.43)	0.30* (0.16)	0.02 (0.03)	-0.02 (0.10)	0.12*** (0.04)
Control Mean	5.71	1.80	0.92	0.97	0.25
State-year Observations	534	534	533	494	465

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represents the change in substantiated cases of child maltreatment by type per 1,000 state population of children aged 0 to 17. Data are from the 2010-2022 Children’s Bureau reports and include all cases of substantiated child maltreatment. All regressions include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

Table 5: Estimates for effects of RML with recreational dispensaries on unique child victims by age (Children’s Bureau, 2010-2022)

	(1)	(2)	(3)	(4)	(5)
	Ages 0-3	Ages 4-7	Ages 8-11	Ages 12-15	Ages 16-17
Cases (per 1,000)					
<i>RML with dispensaries</i>	0.39** (0.15)	0.26** (0.12)	0.19** (0.09)	0.21** (0.08)	0.11** (0.04)
Control Mean	2.90	2.02	1.65	1.50	0.50
Effect Size	13.4%	12.9%	11.5%	14.0%	22.0%
State-year Observations	534	534	534	534	534

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

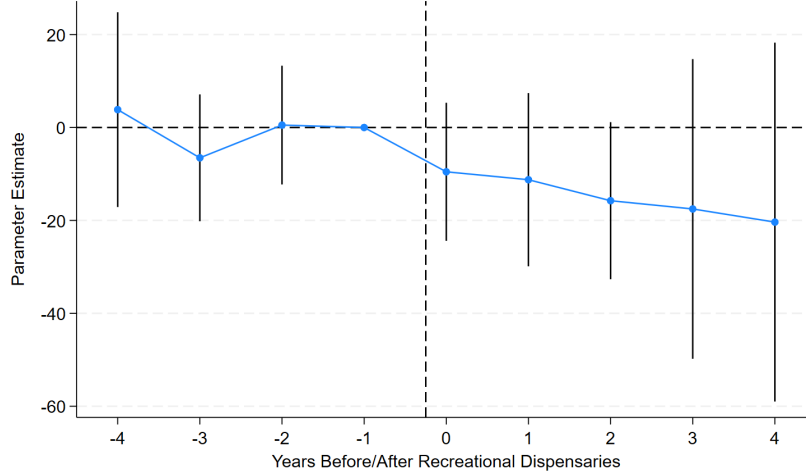
Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represents the change in substantiated cases of child maltreatment by age group per 1,000 state population of children aged 0 to 17. Data are from the 2010-2022 Children’s Bureau reports and include all cases of substantiated child maltreatment. All regressions include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

5.5 RMLs and time use

In Figure 4, I examine a potential mechanism using ATUS data for 2010–2022. The event study plots the effect of RMLs with operating dispensaries on parents’ daily minutes spent caring for children ages 0–17 in the household. Pre-treatment estimates are flat and statistically indistinguishable from zero, both individually and jointly. Following dispensaries opening, parental time in childcare falls and remains lower for at least four years, consistent with a cannabis-induced reduction in day-to-day supervision for a subset of the population. Although none of the post-period estimates are significant at the 5% level, the first three periods are at 10%. Tested jointly, the effect is significant (Table 6) and implies a modest decrease of about 12 minutes per day.

These patterns are consistent with expanded cannabis access leading some adults, plausibly including parents of minors, to reallocate a portion of their day away from hands-on parenting. Although the effect size is small in number of minutes per day, the proportion of the population that is treated (in the increased rate of people using marijuana) is also small. Thus, the effect is carried by a small group of people whose time use could change by several hours each day. This is consistent with changes in parenting behavior that may feasibly result in an increase in neglectful conditions.

Figure 4: Sun & Abraham Event Study, Effect of RML with recreational dispensaries on parental time spent caring for household children (ATUS, 2010-2022)



Notes: The graph contains coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficients represent the change in minutes spent on caring for own children per day by parents of children aged 0 to 17. The regression controls for all covariates, state and time fixed effects. Standard errors are clustered at the state level. Data are from the 2010-2022 ATUS.

Table 6: Summary estimates for effect of RML with recreational dispensaries on parental time use (ATUS, 2010-2022)

(1)	
Panel A: Time Spent Caring for Household Children (minutes per day)	
<i>RML with dispensaries</i>	-12.25** (6.12)
F-test Leads (p-value)	0.30
Control Mean	86.75
Percent Effect	14.1%
State-year Observations	561
Model	Sun & Abraham ¹

* $p < 0.10$, ** $p < .05$, *** $p < .01$.

Notes: Each column represents a separate regression. Results are derived using Sun and Abraham’s (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficients represent the change in minutes spent on caring for own children per day by parents of children aged 0 to 17. Data from the 2010-2022 ATUS. All regressions include state and year fixed effects. Standard errors, clustered at the state level, are in parentheses.

¹ Sun & Abraham point estimates are an average treatment effect derived from a weighted linear combination of estimates from all post-treatment periods. Never-treated units are used for the control group.

It is also possible that as overall rates of use increase with expanded recreational marijuana access, this may raise the rate of individuals developing cannabis use disorder (CUD), which would characteristically present with a chronic shift in time use (Fink et al., 2022; Hoch et al., 2025). When enough time is displaced from prior caregiving activities, children’s needs might not be met as consistently as before the disordered cannabis use, increasing the risk of a CPS investigation resulting in a substantiated allegation of neglect.

The ATUS evidence therefore provides a behavioral bridge between the NSDUH first stage (increased adult marijuana use) and the NCANDS outcome (higher rates of substantiated neglect), and supports the interpretation that routine household behavior is a key channel.

6 Discussion

This paper documents that the final stage of recreational marijuana law implementation, the point at which retail dispensaries open, is associated with higher rates of substantiated child maltreatment, driven by increased rates of neglect. The magnitude is modest in absolute terms, roughly equal to 1.3 additional cases per 1,000 children, but nontrivial relative to a baseline of 8.5 per 1,000. The pattern is not apparent when recreational marijuana law passage is used as the treatment indicator; it becomes visible only once the analysis separates statutory legalization from actual market access.

A first implication is that some prior results in the existing literature - particularly work showing declines in foster care entries for parental substance use or neglect after legalization (Gardner & Osei, 2022) - do not, by themselves, establish that conditions for children have improved. Foster care entries, screened-in reports, and hotline calls may be shaped by evolving professional norms around marijuana, by law enforcement contact, and by agency decisions about whether removal is appropriate. If legalization makes agencies more willing to keep children at home by substituting in-home services for removal in cases involving marijuana, then entries into foster care can fall even while the rate of maltreatment may rise. The results here are consistent with such a substitution: they point to an increase in formal findings of neglect when cannabis access expands.

A second implication is that cannabis policy can affect child welfare through routine household behavior rather than through dramatic child safety incidents. The supplementary time-allocation evidence suggests a channel in which adults reallocate time away from hands-on caregiving once dispensaries open. Accumulated reductions in supervision, responsiveness, or provision of basic care can raise the likelihood that an allegation of neglect is substantiated, particularly in households where parenting deficits are already known to child welfare. This is consistent with the fact that neglect is often tied to cumulative lapses in care, as parents continually fail to meet children’s needs for basic care or hygiene, medical or dental care, educational or mental health support.

This study’s findings appear to show that, alongside a small but statistically significant increase in adult marijuana use after dispensaries open, there is a concurrent reduction in parents’ time spent on childcare. Even though the estimated decline in daily caregiving time is modest on average

(on the order of 10–15 minutes per day), accumulated shortfalls in supervision or routine care can plausibly raise the probability that an allegation is substantiated. This may be especially true for households already known to child welfare agencies.

At the same time, the results should be interpreted with several qualifications. First, although the empirical strategy exploits staggered policy timing and uses an estimator designed for differential adoption, this remains observational evidence. States that are willing to move rapidly from legalization to operational markets may also be states that are expanding surveillance or services for at-risk families. If such policy bundles are not fully captured by the controls, part of the estimated effect could reflect contemporaneous child-welfare reforms. Second, the outcome observed here is an aggregate, state-level measure of substantiated cases. It does not permit identification of which families or which types of neglect are driving the increase aside from the differentiation of medical neglect. Neglect cases linked to inadequate supervision may respond differently to cannabis access than cases tied to parental mental health problems, co-occurring substance use, or economic stress. Third, the analysis does not directly observe adult cannabis consumption for parents involved in substantiations. The mechanism is therefore inferred from population-level shifts in access and from time-use evidence, and should be confirmed with administrative or case-level data where available.

These limitations point to several directions for future work. Linking state-level cannabis policy changes to individual-level or case-level child-welfare data would allow researchers to test whether the increase is concentrated among families with prior CPS history, among very young children, or among cases where substance use was already noted. Researchers should also seek to distinguish between genuine deterioration in caregiving and changes in how caseworkers evaluate marijuana use once it is legal. In addition, comparing states that coupled legalization with clear CPS guidance on parental marijuana use to states that did not would shed light on how much of the observed effect is regulatory rather than behavioral.

Overall, the results suggest that recreational marijuana markets carry a significant child welfare cost. For policymakers, the key lesson is that the child-welfare system should be integrated with cannabis policy in more deliberate ways, such as guidance to CPS workers and parent-targeted education, so that gains from ending prohibition do not come at the expense of children who remain in at-risk homes.

A Appendix Tables

Table A1: State Marijuana Policy Timeline

State	MML Effective Date	RML Effective Date	Recreational Sales Allowed
Alaska	03/04/1999	02/24/2015	10/29/2016
Arkansas	11/09/2016	-	-
Arizona	04/14/2011	11/30/2020	01/22/2021
California	11/06/1996	11/09/2016	01/01/2018
Colorado	06/01/2001	12/10/2012	01/01/2014
Connecticut	08/20/2014	07/01/2021	-
Delaware	06/26/2015	-	-
District of Columbia	07/30/2013	02/26/2015	02/26/2015
Florida	07/26/2016	-	-
Hawaii	12/28/2000	-	-
Illinois	11/09/2015	01/01/2020	01/01/2020
Louisiana	08/06/2019	-	-
Maine	12/22/1999	01/31/2017	10/09/2020
Maryland	12/02/2017	-	-
Massachusetts	01/01/2013	12/15/2016	11/20/2018
Michigan	12/04/2008	12/06/2018	12/01/2019
Minnesota	07/01/2015	-	-
Mississippi	02/02/2022	-	-
Missouri	10/17/2020	12/08/2022	-
Montana	11/02/2004	01/01/2021	01/01/2022
Nevada	10/01/2001	01/01/2017	07/01/2017
New Hampshire	05/01/2016	-	-
New Jersey	12/06/2012	02/22/2021	04/21/2022
New Mexico	07/01/2007	06/29/2021	04/01/2022
New York	01/08/2016	03/31/2021	12/29/2022
North Dakota	03/01/2019	-	-
Ohio	01/16/201	-	-
Oklahoma	07/26/2018	-	-
Oregon	12/03/1998	07/01/2015	10/01/2015
Pennsylvania	01/17/2018	-	-
Rhode Island	01/03/2006	5/25/2022	12/01/2022
South Dakota	07/01/2021	-	-
Utah	03/02/2020	-	-
Vermont	07/01/2004	07/01/2018	10/01/2022
Virginia	10/17/2020	07/01/2021	-
Washington	11/03/1998	12/06/2012	07/08/2014
West Virginia	08/22/2017	-	-

Notes: all original policy dates are drawn directly from (Anderson & Rees, 2023). Updated dates, in bold print, are drawn from the author’s research into legislative changes.

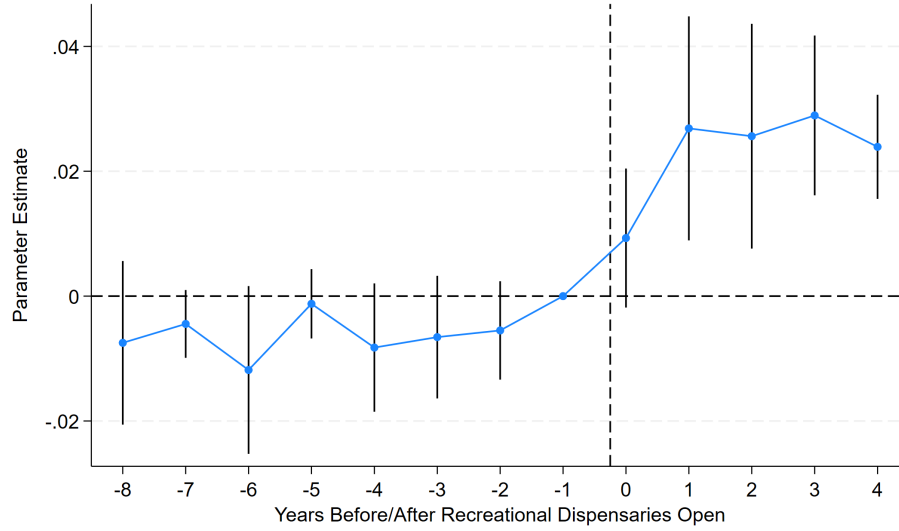
Table A2: Comparison of TWFE and Sun & Abraham Estimators

Concept	What's Included in Estimation	Effect Estimated From	What it Estimates
TWFE (Summary Estimate)	All units. Pre-treatment periods for treated states are included in control mean.	Differences in pre vs. post, across all units (incl. early vs late)	Average Treatment Effect on Treated (ATT)
Sun & Abraham (Event Study)	Select units. Pre-treatment periods are excluded from control mean.	Only post-treatment differences for each cohort vs. never-treated	Event-time-specific CATTs
Sun & Abraham (Summary Estimate)	Average of post-treatment CATTs	Only post-treatment differences for each cohort vs. never-treated	ATT that is consistent with TWFE conceptually, but free from timing bias

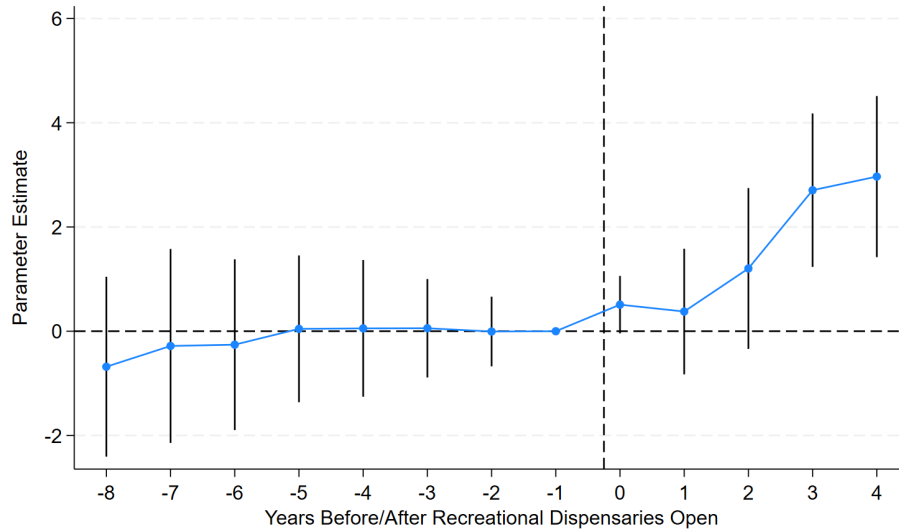
B Appendix Figures

Figure B1: Expanded Event Studies, Primary Outcomes

(a) Any Marijuana Use



(b) Substantiated Child Neglect



Notes: The graphs contain coefficient estimates with 95% confidence intervals for years before and after the change in RML, derived using Sun and Abraham's (2021) dynamic treatment effect estimator to account for timing-induced bias. Coefficient represent (a) the change in the prevalence of marijuana use among adults aged 26+; and (b) the change in substantiated cases of child maltreatment per 1,000 state population of children aged 0-17. The regressions controls for all covariates, state and time fixed effects. Standard errors are clustered at the state level. Data are from the 2010-2019 NSDUH and 2010-2022 NCANDS.

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